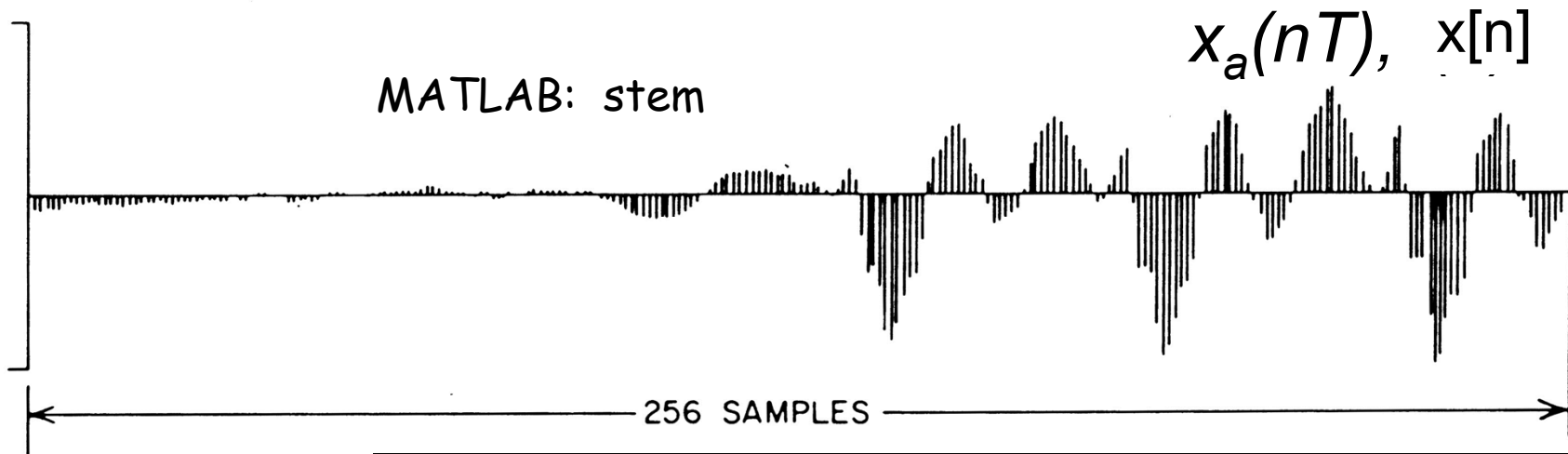
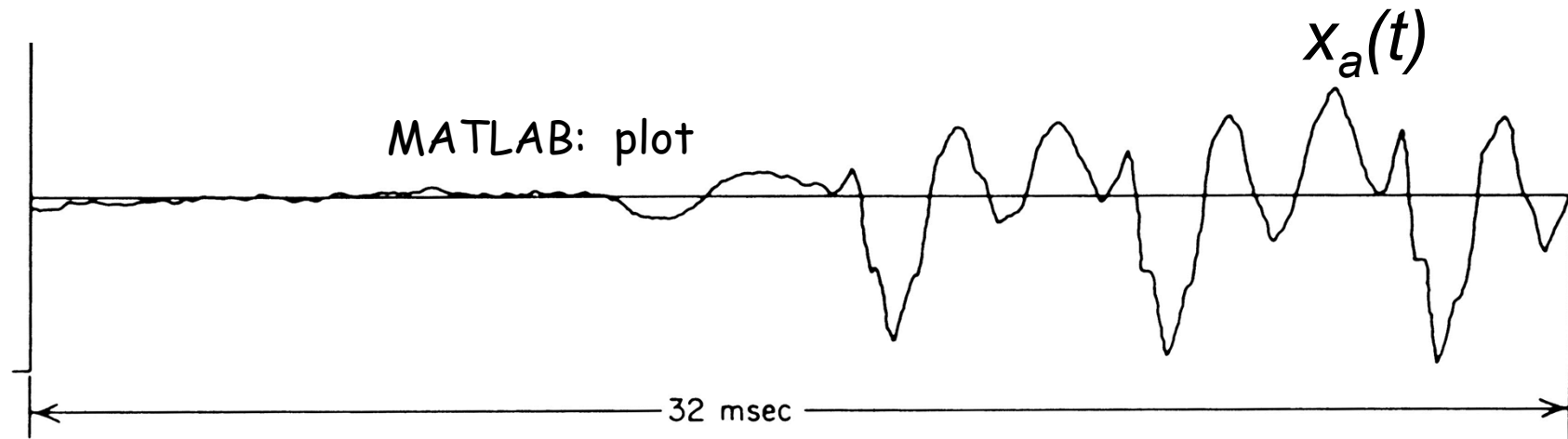


Sampled Speech Waveform

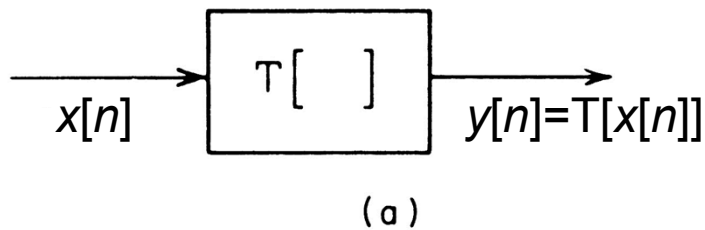


$T=0.125 \text{ msec}, f_s=8 \text{ kHz}$

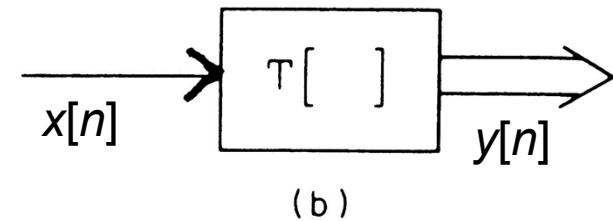
Trap #1: loss of sampling time index

Signal Processing

- Transform digital signal into more desirable form

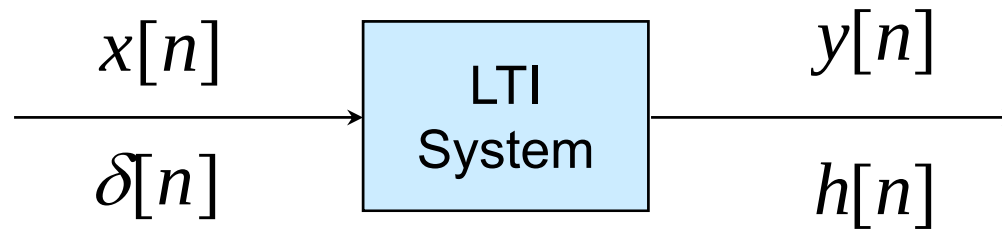


↑
single input—single output



↑
single input—multiple output,
e.g., filter bank analysis,
sinusoidal sum analysis, etc.

LTI Discrete-Time Systems



- Linearity (superposition):

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\}$$

- Time-Invariance (shift-invariance):

$$x_1[n] = x[n - n_d] \Rightarrow y_1[n] = y[n - n_d]$$

- LTI implies discrete convolution:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n] = h[n] * x[n]$$

LTI Discrete-Time Systems

Example:

Is system $y[n] = x[n] + 2x[n + 1] + 3$ linear?

$$x_1[n] \leftrightarrow y_1[n] = x_1[n] + 2x_1[n + 1] + 3$$

$$x_2[n] \leftrightarrow y_2[n] = x_2[n] + 2x_2[n + 1] + 3$$

$$x_1[n] + x_2[n] \leftrightarrow y_3[n] = x_1[n] + x_2[n] + 2x_1[n + 1] + 2x_2[n + 1] + 3 \\ \neq y_1[n] + y_2[n] \Rightarrow \text{Not a linear system!}$$

Is system $y[n] = x[n] + 2x[n + 1] + 3$ time/shift invariant?

$$y[n] = x[n] + 2x[n + 1] + 3$$

$$y[n - n_0] = x[n - n_0] + 2x[n - n_0 + 1] + 3 \Rightarrow \text{System is time invariant!}$$

Is system $y[n] = x[n] + 2x[n + 1] + 3$ causal?

$y[n]$ depends on $x[n + 1]$, a sample in the future

$\Rightarrow$ System is not causal!

LTI Discrete-Time Systems

EXAMPLE The system defined by

$$y(n) = \frac{x^2(n)}{x(n-1)}$$

is not additive because

$$T[x_1(n) + x_2(n)] = \frac{(x_1(n) + x_2(n))^2}{x_1(n-1) + x_2(n-1)}$$

which is not the same as

$$T[x_1(n)] + T[x_2(n)] = \frac{x_1^2(n)}{x_1(n-1)} + \frac{x_2^2(n)}{x_2(n-1)}$$

This system is, however, homogeneous because, for an input $cx(n)$ the output is

$$T[cx(n)] = \frac{(cx(n))^2}{cx(n-1)} = c \frac{x^2(n)}{x(n-1)} = cT[x(n)]$$

LTI Discrete-Time Systems

$$y(n) = x(n) + x^*(n-1)$$

is additive because

$$[x_1(n) + x_2(n)] + [x_1(n-1) + x_2(n-1)]^* = [x_1(n) + x_1^*(n-1)] + [x_2(n) + x_2^*(n-1)]$$

However, this system is not homogeneous because the response to $cx(n)$ is

$$T[cx(n)] = cx(n) + c^*x^*(n-1)$$

which is not the same as

$$cT[x(n)] = cx(n) + cx^*(n-1)$$

LTI Discrete-Time Systems

However, the system described by the equation

$$y(n) = x(n) + x(-n)$$

is shift-varying. To see this, note that the system's response to the input $x(n) = \delta(n)$ is

$$y(n) = \delta(n) + \delta(-n) = 2\delta(n)$$

whereas the response to $x(n - 1) = \delta(n - 1)$ is

$$y'(n) = \delta(n - 1) + \delta(-n - 1)$$

Because this is not the same as $y(n - 1) = 2\delta(n - 1)$, the system is shift-varying.

Convolution Example

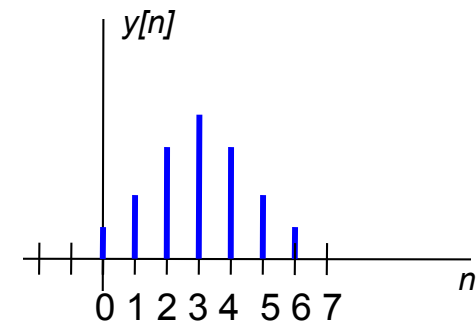
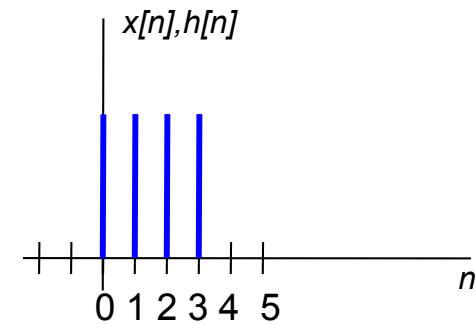
$$x[n] = \begin{cases} 1 & 0 \leq n \leq 3 \\ 0 & \text{otherwise} \end{cases} \quad h[n] = \begin{cases} 1 & 0 \leq n \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

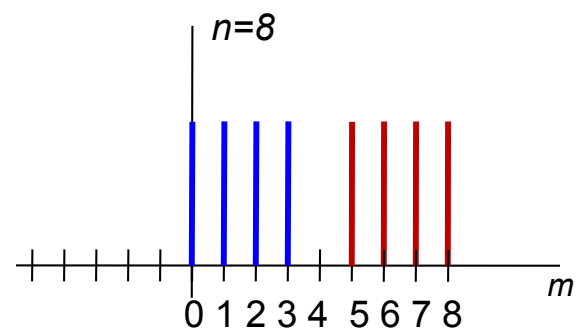
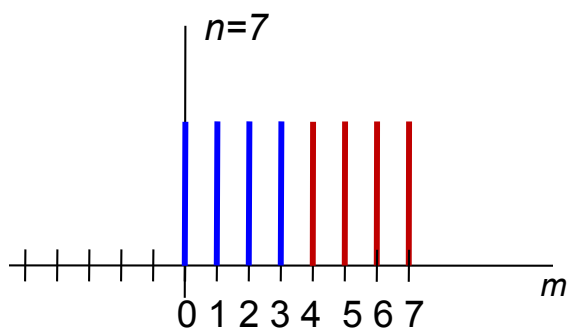
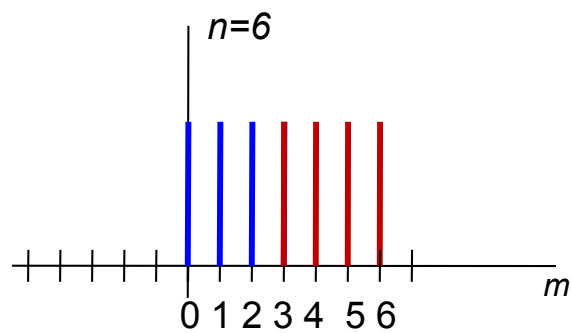
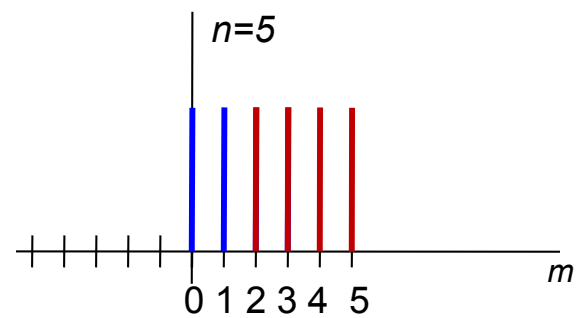
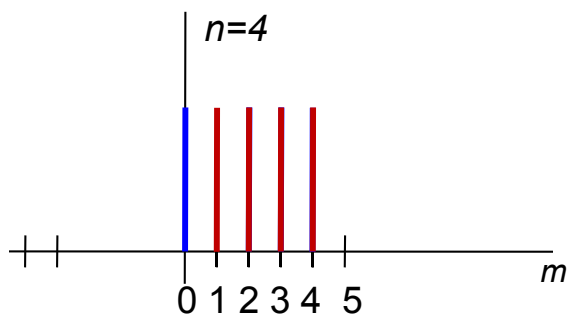
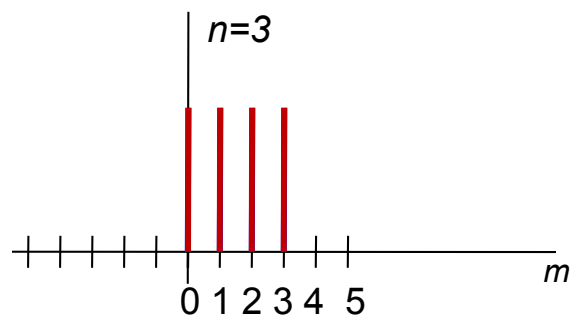
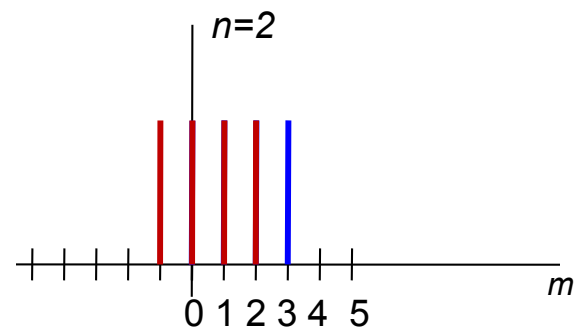
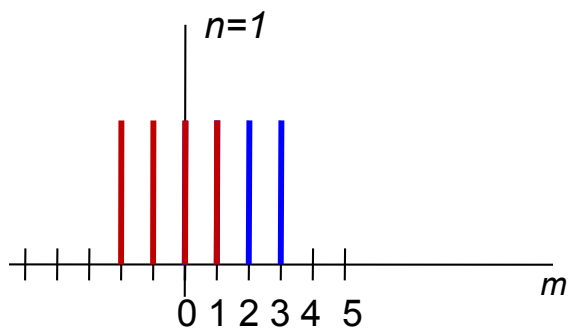
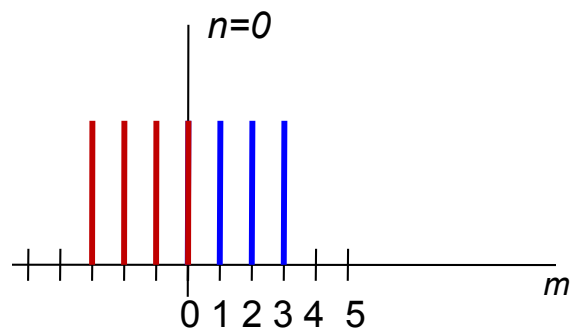
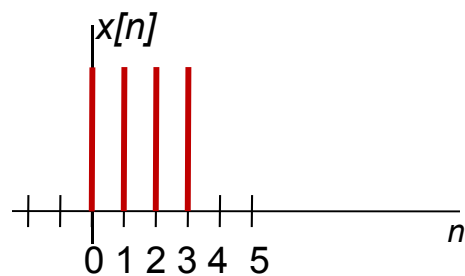
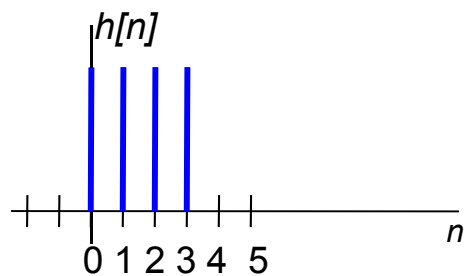
What is $y[n]$ for this system?

Solution :

$$y[n] = x[n] * h[n] = \sum_{m=-\infty}^{\infty} h[m] x[n-m]$$

$$= \begin{cases} \sum_{m=0}^n 1 \cdot 1 = (n+1) & 0 \leq n \leq 3 \\ \sum_{m=n-3}^3 1 \cdot 1 = (7-n) & 4 \leq n \leq 6 \\ 0 & n \leq 0, n \geq 7 \end{cases}$$





Convolution Example

The impulse response of an LTI system is of the form:

$$h[n] = a^n u[n] \quad |a| < 1$$

and the input to the system is of the form:

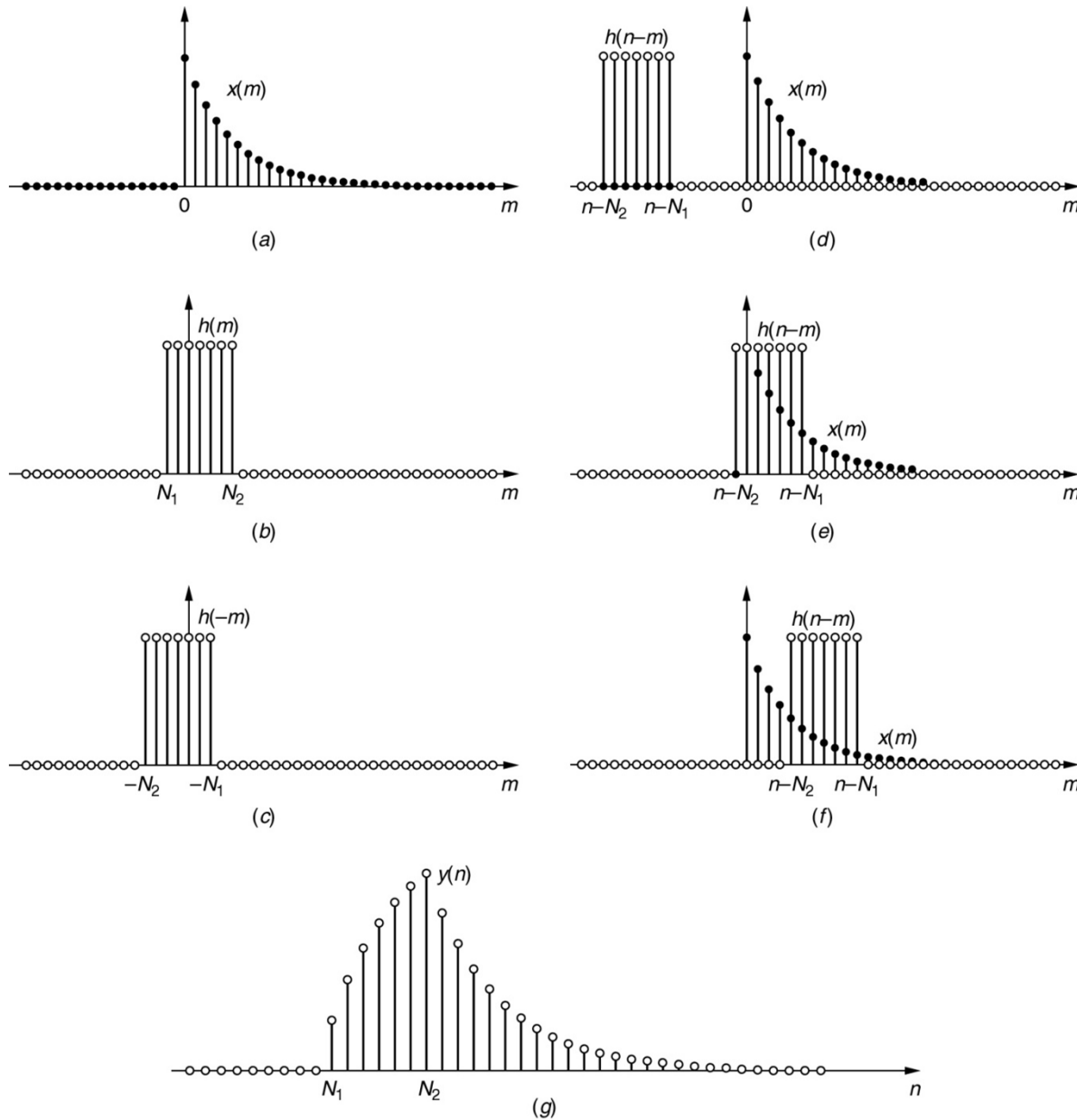
$$x[n] = b^n u[n] \quad |b| < 1, b \neq a$$

Determine the output of the system using the formula for discrete convolution.

SOLUTION:

$$\begin{aligned} y[n] &= \sum_{m=-\infty}^{\infty} a^m u[m] b^{n-m} u[n-m] \\ &= b^n \sum_{m=0}^n a^m b^{-m} u[n] = b^n \sum_{m=0}^n (a/b)^m u[n] \\ &= b^n \left[\frac{1 - (a/b)^{n+1}}{1 - (a/b)} \right] = \left[\frac{b^{n+1} - a^{n+1}}{b - a} \right] u[n] \end{aligned}$$

Convolution Example



Convolution properties

Commutativity:

$$x[n] * h[n] = h[n] * x[n]$$

Associativity:

$$(h_1[n] * h_2[n]) * h_3[n] = h_1[n] * (h_2[n] * h_3[n])$$

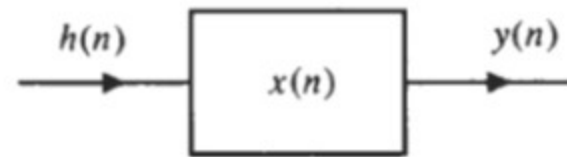
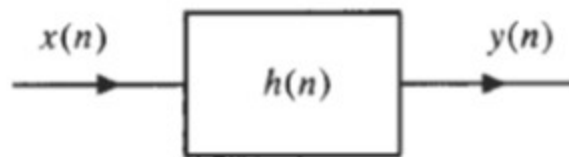
Distributivity:

$$h[n] * (ax_1[n] + bx_2[n]) = a(h[n] * x_1[n]) + b(h[n] * x_2[n])$$

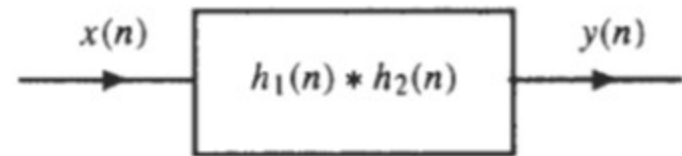
Time reversal:

$$y[-n] = x[-n] * h[-n]$$

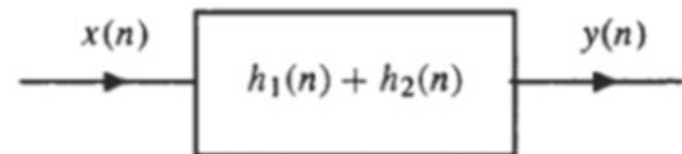
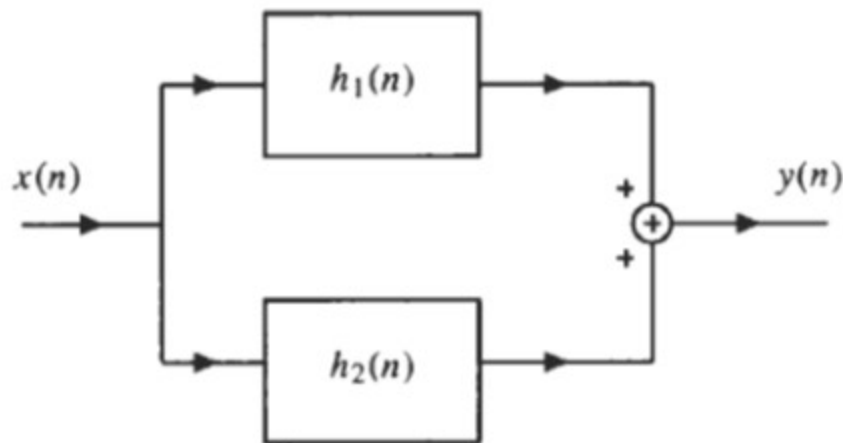
Equivalent LTI Systems



(a) The commutative property.

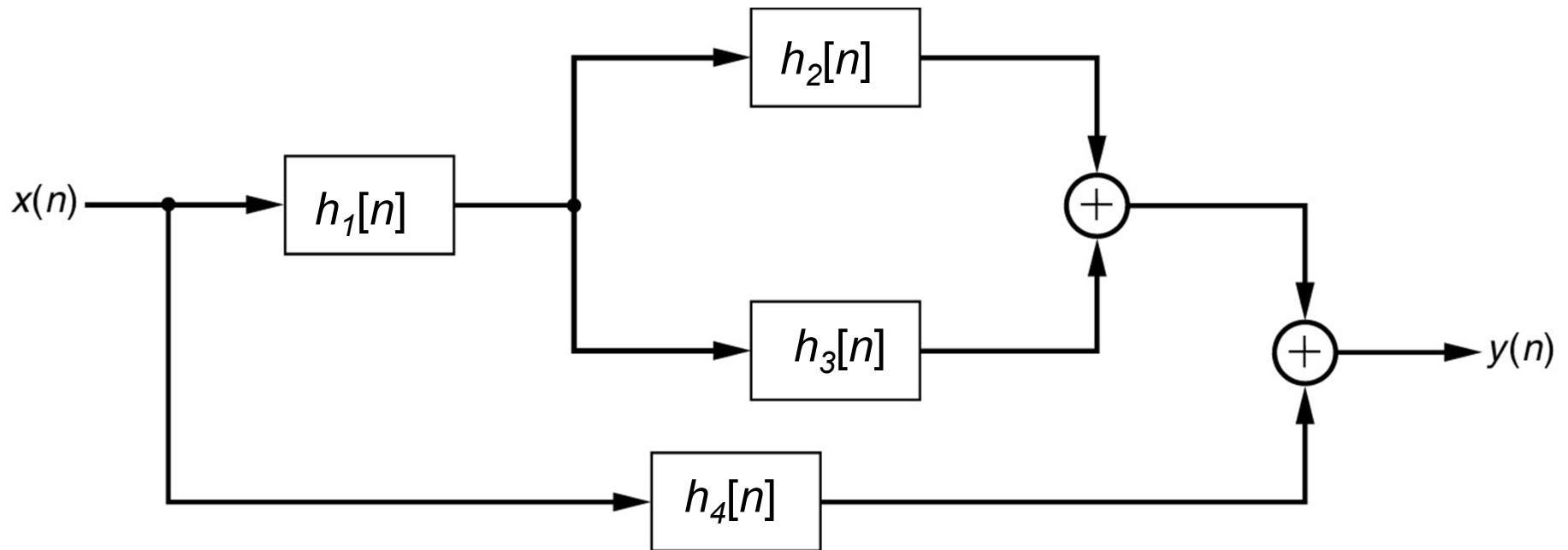


(b) The associative property.



(c) The distributive property.

More Complex Filter Interconnections



$$y[n] = x[n] * h_c[n]$$

$$h_c[n] = h_1[n] * (h_2[n] + h_3[n]) + h_4[n]$$